

SURROUND
Material Carbon Passport

22@ Poblenu Residential Block

Site: 22@ Poblenu, Barcelona, Spain
Building Type: Residential
Stage A1–A3 Embodied Carbon Assessment

1. Material Carbon Passport — Key Metrics

434.43 t	1337 kg/m ²	+457.0%
Total CO ₂ e (tonnes)	Carbon Intensity	vs 240.0 kg/m ² benchmark (above benchmark)

2. Carbon Split by Building Element

Element	CO ₂ e (kg)	CO ₂ e (t)	% of Total
Floor	428,610.0	428.61	98.7%
Wall	4,768.5	4.77	1.1%
Roof	1,047.6	1.05	0.2%
TOTAL	434,426.1	434.43	100%

3. Material Comparative Table — Top Choice per Surface

The table below shows the highest-ranked material option for each building surface, selected by combined score (70% lowest carbon / 30% nearest supplier).

ID	Surface Type	Area (m²)	Material Category	Product Name	Manufacturer	Country	Total CO₂e (kg)	CO₂e per m² (kg/m²)	Distance (km)
1	Floor	120.0	Steel	GreenSteel Structural Section RH150x100	GreenSteel Manufacturing	UK	428610.0	3571.75	16947
2	Roof	120.0	Wood	StructaBoard Timber CLT Panels	EcoTimber Innovations Pty Ltd	Australia	1047.6	8.73	16824
3	Wall	85.0	Concrete	EcoSlab Ready-Mix Concrete	Greencrete Solutions Pty Ltd	Australia	4768.5	56.1	17179

4. Baseline vs Altered Scenario

Comparison of the selected EPD-verified material set against the 22@ Poblenou district average of 207 kg CO₂e/m² and the industry benchmark of 240.0 kg CO₂e/m².

Scenario	Intensity (kg CO₂e/m²)	vs Benchmark
22@ District Average	207	-13.8%
Industry Benchmark	240.0	0.0%
This Design (EPD-verified)	1336.7	+457.0%

Generated by Surround Carbon Pipeline · Stage A1–A3 only · Data sourced from EPD database (2050-materials.com) · Benchmark: 240 kg CO₂e/m² (industry standard)